

Safety Data Sheet



Product name: Vitamin D3 500

Product code: SLDF01

Current version: 3.3, issued: Jan.01,2023 Replaced version: 3.2, issued: Jan.24,2022

Section 1. Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier:

Product name: **Vitamin D3 500**
Product code: SLDF01
Brand: NHU
CAS No.: 67-97-0
EC No.: 200-673-2

1.2 Relevant identified uses of the substance or mixture and uses advised against:

Relevant identified uses of the substance or mixture

Vitamins feed additives.

1.3 Details of the supplier of the safety data sheet:

Company

Zhejiang NHU CO. Ltd

Address

HIGH TECH INDUSTRY ZONE(SECOND),
XINCHANG COUNTY,ZHEJIANG PROVINCE,P.R. OF CHINA

Tel: +86-575-86039932

Fax: +86-575-86165366

E-Mail: sales@cnhu.com

Website: <http://www.cnhu.com/>

1.4 Emergency telephone number:

Please contact the regional NHU representation in your country.

Section 2. Hazard identification

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 (CLP):

Acute Tox, 4; H332

STOT RE 2; H373

For the full text of the H-Statements mentioned in this Section, see Section 16.

Classification according to Hazard Communication Standard (29 CFR 1910.1200)

Specific target organ toxicity after repeated exposure, Category 1

2.2 Label elements

Labelling according to Regulation (EC) No 1272/2008 (CLP Regulation):

Hazard Pictograms

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GHS07



GHS08

Signal word

Warning

Hazardous component(s) to be indicated on label:

Vitamin D3

Hazard statements

H332 Fatal if inhaled.

H373 May cause damage to organs through prolonged or repeated exposure.

Precautionary statements

P260 Do not breathe dust/fume/gas/mist/vapours/spray.

P312 Call a POISON CENTER or doctor/physician if you feel unwell.

P501 Dispose of contents/container in accordance with local and national regulations.

Labelling according to Hazard Communication Standard (29 CFR 1910.1200):

Hazard Pictograms



Signal Word

Danger

Hazard Statements

Causes damage to organs through prolonged or repeated exposure.

Precautionary Statements

Do not breathe dust/fume/gas/mist/vapors/spray.

Wash thoroughly after handling.

Do not eat, drink or smoke when using this product.

Get medical advice/attention if you feel unwell.

Dispose of contents/container in corroding with local regulation.

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2.3 Other hazards

Not available

Section 3.Composition and information on ingredients

3.1 Substances

Not applicable. The product is not a substance.

3.2 Mixtures

Chemical characterization

Vitamin D3 in gelatin.

Hazardous ingredients

No	Substance name			Additional information
	CAS/EC/Index NO.	Classification(EC) 1272/2008(CLP)	Classification HCS 2012	Concentration(%,b.w.)
1	Vitamin D3			
	67-97-0 200-673-2 603-180-00-4	Acute Tox.2*; H330 Acute Tox.3*; H311 Acute Tox.3*; H301 STOT RE 1; H372**	Acute Tox.2 (Inhalation) Acute Tox.3 (Dermal) Acute Tox.3 (Oral) STOT R1	1.25-1.375
2	Sucrose			
	57-50-1 200-334-9 ----	----	----	25.0-35.0
3	Cornstarch			
	9005-25-8 232-679-6 ----	----	----	20.0-35.0

Full text for all R-phrases, H-phrases and EUH-phrases: pls. see section 16

(*, **, ***, ****) Detailed explanation pls. refer to CLP regulation No. 1272/2008, annex VI, 1.2

Additional ingredients

No	Substance name	CAS Number	Concentration(%,b.w.)
1	Gelatin	9000-70-8	10.0-20.0
2	Maltodextrin	9050-36-6	10.0-20.0

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3	BHT	128-37-0	0-1.0
4	Silicon dioxide	7631-86-9	0-1.0

Section 4. First aid measures

4.1 Description of first aid measures:

General information:

If the patient is likely to become unconscious, place and transport in stable sideways position. Remove contaminated clothing immediately and dispose of safely.

In case of inhalation:

Take medical treatment. Ensure supply of fresh air. Breathing with the help of a ventilator bag or ventilator.

In case of eye contact:

Begin with medical treatment. In case of contact with eyes rinse thoroughly with water.

In case of skin contact:

Wash off immediately with soap and water.

In case of ingestion:

Rinse out mouth and give plenty of water to drink. Induce vomiting if patient is conscious, seek medical advice. Call a doctor immediately.

4.2 Most important symptoms and effects, both acute and delayed

No data available.

4.3 Indication of any immediate medical attention and special treatment needed

No data available.

Section 5. Fire-fighting measures

5.1 Extinguishing media:

Suitable extinguishing media :

Water spray jet, alcohol-resistant foam, dry extinguishing powder or carbon dioxide.

Unsuitable extinguishing media:

No data available.

5.2 Special hazards arising from the substance or mixture:

In case of combustion evolution of dangerous gases possible.

5.3 Special protective equipment for fire-fighters:

Use self-contained breathing apparatus.

5.4 Further information:

No data available.

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Section 6. Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures:

For non-emergency personnel

Use personal protective clothing. Ensure adequate ventilation. Avoid dust formation.

For emergency responders

No data available. Personal protective equipment (PPE) - see Section 8.

6.2 Environmental precautions:

Do not discharge into surface waters/groundwater.

6.3 Methods and material for containment and cleaning up:

Take up mechanically and send for disposal.

6.4 Reference to other sections:

See section 13 for information on disposal.

Section 7. Handling and storage

7.1 Precautions for safe handling:

Advice on safe handling

Open and handle container with care. Provide good ventilation of working area (local exhaust ventilation, if necessary). Avoid dust formation.

General protective and hygiene measures

Avoid contact with eyes and skin. Remove soiled or soaked clothing immediately. Keep separated from food-stuffs and feed-stocks. Wash hands before breaks and after work.

7.2 Conditions for safe storage, including any incompatibilities:

Technical measures and storage conditions

Protect from heat and direct sunlight. Keep container tightly closed. Store in a dry place.

Protect from light. Keep in a cool place.

Recommended storage temperature

Store in a clean and dry place.

Requirements for storage rooms and vessels

Containers which are opened must be carefully closed and kept upright to prevent leakage.

Always keep in containers of same material as the original one.

7.3 Specific end use(s):

No data available

Section 8. Exposure controls/Personal protection

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8.1 Control parameters:

INGREDIENT DATA:

Source	Ingredient	TWA	STEL	Peak	Notes
US OSHA Permissible Exposure Levels (PELs)-Table Z1	Sucrose	15 mg/m3/ 5mg/m3	Not available	Not available	Total dust
US ACGIH Threshold Limit Values (TLV)	Sucrose	10 mg/m3	Not available	Not available	TLV®Basis: Dental erosion
US NIOSH Recommended Exposure Limits(RELs)	Sucrose	10 (total), 5 (resp)mg/m3	Not available	Not available	Not available
US OSHA Permissible Exposure Levels (PELs)-Table Z1	Cornstarch	15 mg/m3/ 5mg/m3	Not available	Not available	Total dust
US ACGIH Threshold Limit Values (TLV)	Cornstarch	10 mg/m3	Not available	Not available	TLV®Basis: Dermatitis
US NIOSH Recommended Exposure Limits(RELs)	Cornstarch	10 (total), 5 (resp)mg/m3	Not available	Not available	Not available
US ACGIH Threshold Limit Values (TLV)	BHT	2 mg/m3	Not available	Not available	TLV®Basis: URT irr
US NIOSH Recommended Exposure Limits(RELs)	BHT	10 mg/m3	Not available	Not available	Not available
US OSHA Permissible Exposure Levels (PELs)- Table Z1	Silicon dioxide	Not available	Not available	Not available	See Table Z-3
US OSHA Permissible	Silicon dioxide	Not available	Not available	Not available	See Table Z-3; containing less

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Exposure Levels (PELs)- Table Z1					than 1% crystalline silica
US OSHA Permissible Exposure Levels (PELs)- Table Z3	Silicon dioxide	80/ (%SiO ₂) mg/m ³ / 20 mppcf	Not available	Not available	including natural diatomaceous earth
US NIOSH Recommended Exposure Limits (RELs)	Silicon dioxide	6 mg/m ³	Not available	Not available	Not available

EMERGENCY LIMITS:

Ingredient	TEEL-1	TEEL-2	TEEL-3
Sucrose	30 mg/m ³	980 mg/m ³	5900mg/m ³
Gelatin	0.81 mg/m ³	8.9 mg/m ³	54 mg/m ³
Cornstarch	30 mg/m ³	1200 mg/m ³	1400mg/m ³
BHT	6 mg/m ³	16 mg/m ³	180mg/m ³
Silicon dioxide	6 mg/m ³	6 mg/m ³	6 mg/m ³
Silicon dioxide	6 mg/m ³	6 mg/m ³	630 mg/m ³
Silicon dioxide	18 mg/m ³	200 mg/m ³	1200 mg/m ³
Silicon dioxide	0.07 mg/m ³	0.77 mg/m ³	4.6 mg/m ³
Silicon dioxide	0.3 mg/m ³	0.3 mg/m ³	1.6 mg/m ³
Silicon dioxide	6 mg/m ³	6 mg/m ³	85 mg/m ³
Silicon dioxide	0.9 mg/m ³	9.9 mg/m ³	59 mg/m ³

Ingredient	Original IDLH	Revised IDLH
Vitamin D3	Not Available	Not Available
Sucrose	Not Available	Not Available
Gelatin	Not Available	Not Available
Cornstarch	Not Available	Not Available
Maltodextrin	Not Available	Not Available
BHT	Not Available	Not Available
Silicon dioxide	N.E. mg/m ³ / N.E. ppm	3,000 mg/m ³

8.2 Exposure controls

Appropriate engineering controls:

No data available

Individual protection measures, such as personal protective equipment:

Eye/face protection:

Safety glasses (EN 166).

Hand protection:

Rubber gloves

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Respiratory protection:

In case of dust formation, use a fine dust face mask. Self-contained breathing apparatus.
Short term: filter apparatus, Filter A/P2.

Other

Light protective clothing

Environmental exposure controls:

No data available

Section 9. Physical and chemical properties

9.1 Information on basic physical and chemical properties:

Physical state:

Solid

Form /Colour:

Microcapsule/ Whitelike to yellow-brownish.

Formula:

$C_{27}H_{44}O$

Molecular Weight:

384.7g/mol

Odour:

Slight odor

Odour threshold:

No data available

pH value:

No data available

Melting point/range(°C):

No data available

Boiling point/range(°C):

No data available

Decomposition point/decomposition range:

No data available

Flash point:

Value >93°C

Auto-ignition temperature:

No data available

Flammability(solid, gas):

Not applicable

Lower flammability or explosive limits:

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No data available

Upper flammability or explosive limits:

No data available

Vapour pressure:

No data available

Vapour density:

No data available

Evaporation rate:

No data available

Relative Density:

No data available

Solubility in water:

Insoluble.

Solubility(ies):

No data available

Partition coefficient: n-octanol /water:

No data available

Viscosity:

No data available

9.2 Other information:

Granularity :

100% Through Sieve No.20 (US)

>85.0% Through Sieve No.40 (US)

Loss on drying:

≤5.0%

Bulk density:

No data available

Surface tension:

No data available

Explosive properties:

No data available

Oxidising properties:

No data available

Section 10. Stability and reactivity

10.1 Reactivity:

No data available.

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10.2 Chemical stability:

Stable under normal handling and storage conditions described in Section 7.

10.3 Possibility of hazardous reactions:

No data available.

10.4 Conditions to avoid:

Avoid formation of dust. Heat, naked flames and other ignition sources.

10.5 Incompatible materials:

Strong oxidizing agents, strong bases, strong acids.

10.6 Hazardous decomposition products:

Carbon monoxide and carbon dioxide.

Section 11. Toxicological information

11.1 Information on toxicological effects

Acute oral toxicity (result of the ATE calculation for the mixture)

No	Product Name
1	Vitamin D3 500
Remarks	The result of the applied calculation method according to the European Regulation (EC) 1272/2008 (CLP), Paragraph 3.1.3.6, Part 3 of Annex I is outside the values that imply a classification/labelling of this mixture according to table 3.1.1 defining the respective categories (ATE oral >2000 mg/kg)

Acute oral toxicity

No data available

Acute dermal toxicity (result of the ATE calculation for the mixture)

No	Product Name
1	Vitamin D3 500
Remarks	The result of the applied calculation method according to the European Regulation (EC) 1272/2008 (CLP), Paragraph 3.1.3.6, Part 3 of Annex I is outside the values that imply a classification/labelling of this mixture according to table 3.1.1 defining the respective categories (ATE dermal >2000 mg/kg)

Acute dermal toxicity

No data available

Acute inhalational toxicity (result of the ATE calculation for the mixture)

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No	Product Name
1	Vitamin D3 500
ATE (Mixture)	4.0000
Route of exposure/physical from Method	Dust/mist Calculation method according Regulation (EC) No 1272/2008, (CLP), annex I, part 3, section 3.1.3.6.

Acute inhalational toxicity	
No	Product Name
1	Vitamin D3 500
LC50	0.13-0.38 mg/l
Duration of exposure	4h
Species	rat
Remarks	The values mentioned are those of the active ingredient.

Skin corrosion/irritation
No data available
Serious eye damage/irritation
No data available
Respiratory or skin sensitisation
No data available
Germ cell mutagenicity
No data available
Reproduction toxicity
No data available
Carcinogenicity
No data available
STOT-single exposure
No data available
STOT-repeated exposure
No data available
Aspiration hazard
No data available
Other information
Exercise customary precautions when handling chemicals.

Section 12. Ecological information

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12.1 Toxicity:

Toxicity to fish (acute)
No data available
Toxicity to fish (chronic)
No data available
Toxicity to Daphnia (acute)
No data available
Toxicity to Daphnia (chronic)
No data available
Toxicity to algae (acute)
No data available
Toxicity to algae (chronic)
No data available
Bacteria toxicity
No data available

12.2 Persistence and degradability:

No data available

12.3 Bioaccumulative potential:

No data available

12.4 Mobility in soil:

No data available

12.5 Results of PBT & vPvB assessment:

No data available

12.6 Other adverse effects:

No data available

12.7 Other information:

Ecological data are not available.

Do not discharge product unmonitored into the environment.

Section 13. Disposal considerations

13.1 Waste treatment methods:

Product:

Allocation of a waste code number, according to the European Waste Catalogue, should be carried out in agreement with the regional waste disposal company.

Packaging:

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Residuals must be removed from packaging and when emptied completely disposed of in accordance with the regulations for waste removal. Incompletely emptied packaging must be disposed of in the form of disposal specified by the regional disposer.

Section 14. Transport information

14.1 Transport ADR/RID/ADN

The product is not subject to ADR/RID/AND regulations.

14.2 Transport IMDG

The product is not subject to IMDG regulations.

14.3 Transport ICAO-TI/IATA

The product is not subject to ICAO-TI/IATA regulations.

14.4 Other information

No data available

14.5 Environmental hazards

Information on environmental hazards, if relevant, please see 14.1-14.3.

14.6 Special precautions for user

No data available

14.7 Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not relevant

Section 15. Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

EU regulations

Restriction of occupation

Observe employment restrictions for child bearing mothers and nursing mothers in their latest version.

Council Directive 96/82/EC on the control of major-accident hazards involving dangerous substances

Remarks Annex I, part 2, category 2

US regulations

Vitamin D3 (67-97-0) is found on the following regulatory lists	"US Toxic Substances Control Act (TSCA) - Chemical Substance Inventory" List.
Sucrose (57-50-1) is found on the following regulatory lists	"US - Alaska Limits for Air Contaminants" List. "US - Hawaii Air Contaminant Limits" List.

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	"US - Idaho - Limits for Air Contaminants" List. "US - Wyoming Toxic and Hazardous Substances Table Z1 Limits for Air Contaminants" List. "US Toxic Substances Control Act (TSCA) - Chemical Substance Inventory" List.
Gelatin (9000-70-8) is found on the following regulatory lists	"US Toxic Substances Control Act (TSCA) - Chemical Substance Inventory" List.
Cornstarch(9005-25-8) is found on the following regulatory lists	"US - Hawaii Air Contaminant Limits" List. "US - Idaho - Limits for Air Contaminants" List. "US -Alaska Limits for Air Contaminants" List. "US -Wyoming Toxic and Hazardous Substances Table Z1 Limits for Air Contaminants" List. "US Toxic Substances Control Act (TSCA) -Chemical Substance Inventory" List.
Maltodextrin(9050-36-6) is found on the following regulatory lists	"US - Alaska Limits for Air Contaminants" List. "US - Hawaii Air Contaminant Limits" List. "US - Idaho - Limits for Air Contaminants" List. "US - Wyoming Toxic and Hazardous Substances Table Z1 Limits for Air Contaminants" List. "US Toxic Substances Control Act (TSCA) - Chemical Substance Inventory" List.
BHT (128-37-0) is found on the following regulatory lists	"US - Alaska Limits for Air Contaminants" List. "US - Hawaii Air Contaminant Limits" List. "US Toxic Substances Control Act (TSCA) -Chemical Substance Inventory" List.
Silicon dioxide (7631-86-9) is found on the following regulatory lists	"US - Idaho - Toxic and Hazardous Substances-Mineral Dust" List. "US - Hawaii Air Contaminant Limits" List. "US - Wyoming Toxic and Hazardous Substances Table Z-3 Mineral Dusts" List. "US - Idaho - Limits for Air Contaminants" List. "US - Washington Toxic air pollutants and their ASIL, SQER and de minimis emission values" List. "US - Alaska Limits for Air Contaminants" List. "US Toxic Substances Control Act (TSCA) – Chemical Substance Inventory" List.

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15.2 Chemical safety assessment

No data available.

Section 16. Other information

16.1 Full text of the R-, H- and EUH- phrases drawn up in sections 2 and 3 (provided not already drawn up in these sections)

Acute Tox.	Acute toxicity
H301	Toxic if swallowed.
H311	Toxic in contact with skin.
H330	Fatal if inhaled.
H372	Causes damage to organs through prolonged or repeated exposure.
STOT RE	Specific target organ toxicity – repeated exposure.

16.2 Other information:

HMIS® ratings

Health: 3
Flammability: 1
Physical hazard: 0

NFPA ratings

Health: 3
Flammability: 1
Instability: 0

16.3 Further information

The above information is believed to be correct but does not purport to be all inclusive and shall be used only as a guide. This safety data sheet should be used in conjunction with technical sheets. It does not replace them. The information is based on our current knowledge however it does not represent a guarantee of product properties nor does it create any legal obligation. NHU Corporation shall not be held liable for any damage resulting from handling or from contact with the above product.